

## Assessment 4.1: Home Safety and Conservation Quiz

**Unit:** Unit 4, Spaces We Live In (Environmental Design and Management) **Topic:** 4.1 Safe, Healthy, Sustainable Homes (Lessons 4.1 to 4.5) **When:** The last 15 minutes of Lesson 4.5 (day 97), starting at minute 23 of the period. Step 4 of that lesson's procedure holds the slot. **Category:** Topic assessments (25 percent). One retake after a review; best score counts. The retake review runs during Lesson 4.7 Day 1's team sort for students who need it. **Points:** 30 **Allowed:** a calculator. Nothing else. The unit word bank on the wall stays up. **Time:** 15 minutes. Extended time: finish in the first ten minutes of Lesson 4.6 or in a support period.

## Home Safety and Conservation Quiz

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

**Directions:** Twelve items. Items 1 to 4 are matching; write the letter. Items 5 to 10 are short answer; a phrase is enough unless it says “sentence.” Item 11 is a scenario; write two or three sentences. Item 12 is a sort. Show your math on items 8 and 9. You may use a calculator.

### Part A: Matching (items 1 to 4)

Match each hazard to the room where it is most common. Write the letter. Each room is used once.

| Item     | Hazard                                             |   | Room     |
|----------|----------------------------------------------------|---|----------|
| 1. _____ | Gasoline stored next to the water heater           | A | Bathroom |
| 2. _____ | A pot handle turned out over the edge of the stove | B | Stairs   |
| 3. _____ | A hair dryer plugged in with the cord in the sink  | C | Garage   |
| 4. _____ | A loose rug at the top with a curled corner        | D | Kitchen  |

### Part B: Short answer (items 5 to 10)

5. Where do smoke detectors go in a home, and how often do you test them? (two parts)



6. Why does a home need a carbon monoxide (CO) detector even if it has a smoke detector? One sentence.

7. A breaker trips. Write the two moves to reset it, in order. Then: it trips again right away. What do you do?

Move 1: \_\_\_\_\_ Move 2: \_\_\_\_\_ If it trips again: \_\_\_\_\_

8. Energy math. A 100 W bulb runs 4 hours a day for 365 days. Electricity costs \$0.25 per kWh.

$100 \times 4 \times 365 =$  \_\_\_\_\_ Wh. Divide by 1,000 = \_\_\_\_\_ kWh. Times \$0.25 = \$ \_\_\_\_\_ a year.

9. Water math. A faucet drips once a second, about 5 gallons a day. How many gallons in a year? A bathtub holds about 60 gallons. About how many bathtubs is that?

$5 \times 365 =$  \_\_\_\_\_ gallons. That is about \_\_\_\_\_ bathtubs.

10. A bathroom sink drains slowly. Write the first two things you do before you ever reach for a chemical drain cleaner.

First: \_\_\_\_\_ Second: \_\_\_\_\_

### Part C: Scenario (item 11)

11. The Ferreira-Lund family lives in a two-story house: two parents, a 3-year-old, and a grandfather who uses a walker and sleeps on the second floor. Write their family emergency plan in two or three sentences. Your plan must include: two ways out of the grandfather's bedroom, one meeting spot that he can reach, who carries the 3-year-old, and the two numbers the family should know.





**Part D: Sort (item 12)**

12. Put each item in the bin that does the least harm. Write the letter of the bin next to each item.

Bins: R1 = Reduce (do not bring it in), R2 = Reuse, R3 = Recycle, C = Compost

| Item                                                 | Bin |
|------------------------------------------------------|-----|
| A banana peel                                        |     |
| A glass jar                                          |     |
| A stack of junk mail                                 |     |
| A plastic water bottle you have not bought yet       |     |
| A clean cardboard box you no longer need for storage |     |
| A T-shirt with a hole                                |     |

**Grade 8 stretch (no points):** A low-flow showerhead costs \$15 and saves about \$70 a year. How many months until it pays for itself? \_\_\_\_\_

## Teacher key

Total 30 points.

| Item | Answer                                                                                                                                                                                 | Points | Standard line checked                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------|
| 1    | C (Garage)                                                                                                                                                                             | 1      | EDM 1. b); HSW 3. c)                       |
| 2    | D (Kitchen)                                                                                                                                                                            | 1      | EDM 1. b); HSW 3. c)                       |
| 3    | A (Bathroom)                                                                                                                                                                           | 1      | EDM 1. b); HSW 3. c)                       |
| 4    | B (Stairs)                                                                                                                                                                             | 1      | EDM 1. b); HSW 3. c)                       |
| 5    | On every level of the home and in or near every sleeping area (1 point); test monthly (1 point). Accept “every floor and every bedroom.”                                               | 2      | EDM 1. b); Std 2 PI B                      |
| 6    | Carbon monoxide has no smell (and no color), so nobody can notice it without a detector; a smoke detector does not sense CO. Accept any sentence with “no smell” or “cannot smell it.” | 2      | EDM 1. b); HSW 3. c)                       |
| 7    | Move 1: push it fully OFF (1). Move 2: push it ON (1). If it trips again: stop, unplug what you turned on, tell an adult; do not keep resetting it (1).                                | 3      | EDM 1. b); HSW 4. c); PSI 3. a) c., d., e. |
| 8    | 146,000 Wh (1); 146 kWh (1); \$36.50 (1). Accept rounding to \$36 or \$37. If the divide step is missed but the method is otherwise                                                    | 3      | EDM 1. d); SUS 3. e); National FCS 3.4.3   |



|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |                                                   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------|
|    | right, 1 point total.                                                                                                                                                                                                                                                                                                                                                                                                                                |   |                                                   |
| 9  | 1,825 gallons (2);<br>about 30 bathtubs<br>(1). Accept 30 or 31.                                                                                                                                                                                                                                                                                                                                                                                     | 3 | EDM 1. d); SUS 3. e);<br>National FCS 3.4.3       |
| 10 | First: pull out the<br>hair or gunk with a<br>plastic drain tool (or<br>by hand with<br>gloves) (1). Second:<br>plunge with water<br>over the cup, then<br>hot water (1). Order<br>can be swapped if<br>both are named.<br>“Never mix<br>chemicals” as a<br>bonus note, no extra<br>points.                                                                                                                                                          | 2 | EDM 1. b); PSI 3. a)<br>d.; HSW 4. c)             |
| 11 | Two ways out of the<br>grandfather’s room:<br>the door to the hall<br>and the window (or<br>a plan to wait at the<br>window with the<br>door closed) (1). A<br>meeting spot he can<br>reach with a walker:<br>the mailbox, a<br>neighbor’s front<br>step, a fixed object<br>close by; not the<br>driveway (1). A<br>parent carries the 3-<br>year-old (1). 911<br>and Poison Control<br>1-800-222-1222 (1).<br>Written as a plan,<br>not a list (1). | 5 | EDM 1. c); HSW 3.<br>e); HSW 4. g); Std 2<br>PI B |
| 12 | Banana peel C; glass<br>jar R2; junk mail R3<br>(accept R1 with<br>“sign off the list”);<br>water bottle not yet<br>bought R1; clean                                                                                                                                                                                                                                                                                                                 | 6 | EDM 1. d); SUS 2. c);<br>National FCS 2.2.3       |

|  |                                                                                |  |  |
|--|--------------------------------------------------------------------------------|--|--|
|  | cardboard box R3<br>(accept R2); T-shirt<br>with a hole R2. One<br>point each. |  |  |
|--|--------------------------------------------------------------------------------|--|--|

Stretch:  $\$15 / \$70 = 0.21$  years, about 2.6 months; accept “about 3 months.” No points.

**Scoring bands:** 27 to 30 got it; 21 to 26 approaching, no retake needed unless the student wants one; 20 or below, retake after the review. The review for retakes runs during Lesson 4.7 Day 1’s team sort and covers items 7, 8, and 11, which are the ones most often missed.

**Reading the results:** if more than a third of the class misses item 8 on the divide step, run one more bulb problem as the Lesson 4.6 do now before the design topic starts. If item 11 is weak, the Design a Room client brief in Lesson 4.11 should ask every client’s plan for the same three parts.